

SKYG

ARDUPILOT CONFIG

SKYG SPEED 90

SpeedyBee F405 WING MINI (ArduPilot target: SpeedyBeeF405WING) · ArduPlane 4.5 or later

OVERVIEW



Starting ArduPlane configuration for SkyG Speed 90 on the SpeedyBee F405 WING MINI (firmware target SpeedyBeeF405WING, ArduPlane 4.5 or later). Load the .param file in Mission Planner (Config > Full Parameter List > Load from file), then work through the bench checks.

Flight controller: on the wing-bay floor, arrow forward; servo BEC jumper set to 6 V for the ES9052MD servos.

WARNINGS

- These are starting values, not a tested setup. Verify every output, direction and endpoint on the bench before the first flight.
- Never raise THR_MAX: it protects the prop's rpm rating.
- No airspeed sensor: the throttle limits do not stop a dive from building speed. Fly stage runs level.
- SkyG Speed 90 was designed around an rpm governor at 29,500 rpm. ArduPilot cannot run the governor script on an F405 board. Until a governor is fitted (H7 flight controller or a governed ESC), THR_MAX stays at 70 % and the stages that need more throttle are not flown (see the stage table).

FRAME, ORIENTATION AND LOGGING

Board mounted on the wing-bay floor, arrow forward, components up (build guide). Log to the microSD card.

PARAMETER	VALUE	NOTE
AHRS_ORIENTATION	0	FC arrow forward, upright. Change only if the board is mounted otherwise.
LOG_BACKEND_TYPE	1	Logs to the microSD card.
LOG_DISARMED	0	No logging while disarmed.

OUTPUTS

S1 drives the motor with bidirectional DShot (rpm telemetry for the prop-rpm check). S1 and S2 form one timer group: S2 stays unused. S3 drives the single aileron servo (both ailerons through the torque rods). S4 and S5 drive the left and right tail panels as a V-tail mix.

Servo endpoints are calculated for the Emax ES9052MD (about 11.1 us per degree) and the linkages: aileron 2.79:1 (12 mm horn, 4.3 mm arm hole), tail 1.19:1 (8 mm horn, 6.7 mm arm hole). They are starting points: set the final values by measuring the deflections on the bench.

PARAMETER	VALUE	NOTE
SERV01_FUNCTION	70	S1: throttle (motor ESC).
SERV02_FUNCTION	0	S2: unused (shares the DShot timer group with S1).
SERV03_FUNCTION	4	S3: aileron (one servo drives both ailerons).
SERV04_FUNCTION	79	S4: V-tail left (left tail panel).
SERV05_FUNCTION	80	S5: V-tail right (right tail panel).
SERV03_TRIM	1500	Aileron neutral. The 3 deg reflex is set mechanically on each link, not here (one servo moves the ailerons in opposite directions).
SERV03_MIN	1178	About 10 deg aileron (servo 29 deg at 2.79:1). Verify: about 3.5 mm at the aileron trailing edge.
SERV03_MAX	1822	As SERV03_MIN. Linkage limit +-20 deg aileron = +-807 us: never set beyond it.
SERV04_TRIM	1500	Left panel neutral; trim mechanically first (collar on the shaft flat).

OUTPUTS

PARAMETER	VALUE	NOTE
SERV04_MIN	1232	+20 deg panel travel at 1.19:1 (servo 24 deg). Verify by measurement.
SERV04_MAX	1768	As SERVO4_MIN.
SERV05_TRIM	1500	Right panel neutral.
SERV05_MIN	1232	As SERVO4_MIN.
SERV05_MAX	1768	As SERVO4_MIN.
MIXING_GAIN	0.5	V-tail mix: full pitch alone gives half the panel travel (about +/-10 deg), pitch plus yaw can reach the +/-20 deg endpoints.
SERV0_RATE	50	Servo update rate (Hz) for the PWM groups; the ES9052MD is a digital servo and also accepts higher rates.

MOTOR, ESC AND RPM TELEMETRY

Bidirectional DShot gives motor rpm in the logs: check on the thrust stand that the unloaded prop stays below 30,000 rpm on a full pack.

SERVO_DSHOT_ESC and the protocol depend on the ESC firmware (AM32 or BLHeli_32): verify on the bench.

PARAMETER	VALUE	NOTE
SERVO_BLH_MASK	1	DShot on output 1.
SERVO_BLH_0TYPE	5	DShot300. Verify the ESC accepts it.
SERVO_BLH_BDMASK	1	Bidirectional DShot (rpm telemetry) on output 1.
SERVO_BLH_POLES	14	Motor magnet poles (12N14P 2207 motors). Verify for your motor.
SERVO_DSHOT_ESC	1	BLHeli_32 / AM32 type ESC. Verify against your firmware version's list.
RPM1_TYPE	5	RPM sensor from ESC telemetry (logged).
RPM1_ESC_MASK	1	Motor on output 1.

THROTTLE

THR_MAX is the prop-rating cap: no governor on the F405 yet: $30,000 / (2550 \text{ KV} \times 16.8 \text{ V}) = 70 \%$ keeps an unloaded prop on a full pack at its rating. Raise only when an rpm governor is fitted and proven on the thrust stand.

Speed stages are held by a transmitter throttle limit on a switch (LAUNCH / STAGE, table below), not by the flight controller. ArduPilot clips the throttle at THR_MAX in the stabilised modes; check on the bench that MANUAL is clipped too, otherwise fly the stages in FBWA.

PARAMETER	VALUE	NOTE
THR_MAX	70	no governor on the F405 yet: $30,000 / (2550 \text{ KV} \times 16.8 \text{ V}) = 70 \%$ keeps an unloaded prop on a full pack at its rating. Raise only when an rpm governor is fitted and proven on the thrust stand.
THR_MIN	0	Motor stops at zero throttle in flight (gliding, flutter abort).
THR_SLEWRATE	100	Throttle slew limit (% per second).
TRIM_THROTTLE	35	Cruise throttle guess for the synthetic-airspeed modes. Set from a log of level flight.

NO AIRSPEED SENSOR

No airspeed sensor is fitted. ArduPilot uses a synthetic airspeed from GPS, wind estimate and throttle; FBWA and MANUAL do not need it.

Stall is about 13.0 m/s; AIRSPEED_MIN keeps 25 % margin. AIRSPEED_MAX matches the first speed stage and is raised with the stages.

PARAMETER	VALUE	NOTE
ARSPD_TYPE	0	No airspeed sensor.
ARSPD_USE	0	Do not use an airspeed sensor.
AIRSPEED_MIN	17	m/s; 1.25 x the 13.0 m/s stall.
AIRSPEED_CRUISE	20	m/s; cruise target for CRUISE / FBWB / RTL.
AIRSPEED_MAX	30	m/s; stage 1. Raise with each stage the test plan clears.

ATTITUDE LIMITS, PITCH RATE AND YAW DAMPER

Pitch rate is limited per speed stage so a full pull stays under the 6 g limit (table below); the file holds stage 1 (94 deg/s). Lower it before each faster stage.

The flight-dynamics model finds the Dutch roll about neutral at 13 m/s, so the yaw damper is required. Start low and tune in flight.

PARAMETER	VALUE	NOTE
PTCH_RATE_MAX	94	deg/s, stage 1 (30 m/s). See the stage table.
ROLL_LIMIT_DEG	55	Bank limit in the stabilised modes.
PTCH_LIM_MAX_DEG	25	Nose-up limit.
PTCH_LIM_MIN_DEG	-25	Nose-down limit (no steep powered dives: speed builds fast).
YAW2SRV_DAMP	0.15	Yaw damper gain, starting point. Increase in small steps if the tail wags.
KFF_RDDRMIX	0.5	Rudder (V-tail yaw) mixed with aileron for coordinated turns.

RECEIVER, MODES AND ARMING

RadioMaster RP1 V2 ELRS receiver on UART1 (TX1/RX1) with CRSF. Arm with a switch on channel 5; flight modes on channel 6.

Hand launch in FBWA with the LAUNCH throttle position; no auto-takeoff is needed.

PARAMETER	VALUE	NOTE
SERIAL1_PROTOCOL	23	RCIN (CRSF/ELRS) on UART1.
RC_OPTIONS	8192	Bit 13: 420k baud for ELRS.
RC5_OPTION	153	Channel 5 arms and disarms.
ARMING_RUDDER	0	No stick arming (switch only).
ARMING_CHECK	1	All pre-arm checks.
FLTMODE_CH	6	Flight-mode channel.
FLTMODE1	5	FBWA (launch, stage runs).
FLTMODE2	5	FBWA.
FLTMODE3	0	MANUAL.

RECEIVER, MODES AND ARMING

PARAMETER	VALUE	NOTE
FLTMODE4	0	MANUAL.
FLTMODE5	11	RTL.
FLTMODE6	11	RTL.

GPS AND COMPASS

TBS M10Q on UART3 (TX3/RX3, the GPS pads) with its QMC5883 compass on I2C (SDA/SCL). The GPS sits behind the battery: its compass is 3-6 cm from the battery leads.

Either calibrate and run compass-motor interference compensation (compassmot) on the thrust stand, or disable the compass and fly on GPS heading (COMPASS_USE 0).

PARAMETER	VALUE	NOTE
SERIAL3_PROTOCOL	5	GPS on UART3.
GPS_TYPE	1	Auto (u-blox M10). Named GPS1_TYPE in ArduPlane 4.6+.
COMPASS_ENABLE	1	Use the M10Q's compass (set 0 to fly without).
COMPASS_AUTODEC	1	Automatic declination.
COMPASS_AUTO_ROT	2	Detect the compass orientation during calibration. Verify the result against the GPS's arrow.

BATTERY MONITOR AND FAILSAFES

WING MINI PDB sensors: voltage divider 1K:10K, current 50 A/V. Calibrate both against a meter.

Radio failsafe: RTL after 5 s without signal. Low battery: RTL.

PARAMETER	VALUE	NOTE
BATT_MONITOR	4	Analog voltage and current.
BATT_VOLT_PIN	10	PDB voltage.
BATT_CURR_PIN	11	PDB current.
BATT_VOLT_MULT	11.05	Calibrate against a meter.
BATT_AMP_PERVLT	50	Calibrate against a meter or a known charge.
BATT_CAPACITY	650	mAh (4S 650).
BATT_LOW_VOLT	13.6	3.4 V per cell under load.
BATT_LOW_MAH	160	About 25 % left.
BATT_FS_LOW_ACT	1	RTL on low battery.

BATTERY MONITOR AND FAILSAFES

PARAMETER	VALUE	NOTE
THR_FAILSAFE	1	Radio failsafe on.
FS_SHORT_ACTN	0	Short failsafe: circle (default).
FS_LONG_ACTN	1	Long failsafe: RTL.
FS_LONG_TIMEOUT	5	Seconds.

OSD AND WIRELESS TELEMETRY

Analog OSD (board OSD chip) with ground speed, throttle, current, used capacity and voltage. ArduPilot has no ground-speed alarm: watch the ground-speed number against the stage limit and judge speed from two-way passes.

DJI O4 noses: switch to MSP DisplayPort on UART5 (SERIAL5_PROTOCOL 42, OSD_TYPE 5).

PARAMETER	VALUE	NOTE
OSD_TYPE	1	Analog OSD. Use 5 (MSP DisplayPort) with the DJI O4 noses.
OSD1_ENABLE	1	Screen 1.
OSD1_GSPEED_EN	1	GPS ground speed.
OSD1_THROTTLE_EN	1	Throttle %.
OSD1_CURRENT_EN	1	Current.
OSD1_BATUSED_EN	1	Used capacity (mAh).
OSD1_BAT_VOLT_EN	1	Battery voltage.
OSD_W_BATVOLT	14.0	Battery voltage warning (V).
SERIAL6_PROTOCOL	2	MAVLink2 on UART6 (wireless board).

OSD AND WIRELESS TELEMETRY

PARAMETER	VALUE	NOTE
SERIAL6_BAUD	115	115200 baud.

BENCH CHECKS BEFORE FLIGHT

- Props off. Power from a full pack; check the battery voltage and current readings against a meter.
- Control directions in MANUAL: right stick -> right aileron trailing edge up, left down. Pull -> both tail-panel trailing edges up. Right rudder -> left panel trailing edge down, right panel up (inverted V, panels below the fuselage). Fix with SERVOx_REVERSED.
- Measure deflections with a protractor: aileron about +/-10 deg, tail panels +/-20 deg at the endpoints; adjust SERVOx_MIN/MAX. Nothing binds or buzzes at the ends.
- FBWA: roll and pitch the airframe by hand; the surfaces must oppose the motion.
- Thrust stand: THR_MAX 70 %; logged rpm with the prop unloaded on a full pack below 30,000; ESC current within its rating.
- Transmitter: LAUNCH / STAGE switch set to the stage limit; confirm in the servo-output screen that full stick on STAGE gives the stage %.
- Radio failsafe: switch the transmitter off with the motor at low throttle (props off): motor stops or RTL behaviour as set.
- Compass: calibrate, then compassmot on the thrust stand (or COMPASS_USE 0).
- GPS: 3D fix and the OSD shows ground speed.